

Cassie Charlie

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SUMMARY

Computer Science graduate, Academic Excellence Award Recipient (CGPA: 4.08/4.5), and NSTC grant recipient specializing in Generative Speech AI, Diffusion Architectures, and Audio Processing. Seeking an AI engineering role to deploy advanced agentic and deep learning solutions.

EDUCATION

National Dong Hwa University (國立東華大學)
Bachelor of Computer Science and Information Engineering

Hualien, Taiwan
Graduation Date: Jun 2026

WORK EXPERIENCE

National Dong Hwa University
Teaching Assistant | Introductory Machine Learning

Hualien, Taiwan
Oct 2025 - Jan 2026

- Managed classroom logistics for 45 students including equipment setup, attendance tracking, and hands-on ML session support.
- Developed homework solutions and grading rubrics; evaluated 150+ assignments and exams with consistent feedback and fair assessment.
- Conducted office hours and review sessions, helping students debug Python code and understand core ML concepts.

National Dong Hwa University
Research Assistant at Prof. Wen-Chieh Fang's Lab

Hualien, Taiwan
Jun 2025 - Nov 2025

- Optimized diffusion-based speech enhancement via ablation studies and hyperparameter tuning, achieving 26.3% CSIG improvement and 11% STOI gain over baseline.
- Benchmarked architectural variants across 6 evaluation metrics (PESQ, CSIG, CBAK, COVL, SSNR, STOI), identifying optimal conditioning strategies for audio deepfake enhancement.
- Developed chunked inference pipeline with overlap blending, enabling 94% ASV pass rate while maintaining audio quality for full-length samples.

PROJECT EXPERIENCE

National Science and Technology Council
EmoScale | NSTC-Funded Undergraduate Research

Hualien, Taiwan
Jul 2025 - Mar 2026

- Designed intensity learning framework using RankNet on prosodic features (OpenSMILE), achieving 92.45% validation accuracy in predicting relative emotional intensity without manual labels.
- Architected embedding extraction pipeline (HuBERT content, ECAPA-TDNN speaker, Wav2Vec2 emotion) supporting diffusion decoder training for 6 speakers with zero-shot evaluation on 2 unseen speakers.
- Validated intensity controllability via RMSE analysis across scales 0.0-1.0 (increase from 0.000 to 0.0137), confirming model's responsiveness to conditioning signals.

SKILLS & LANGUAGES

- Technical Skills**
- Languages: Python, C/C++, SQL
 - Frameworks & Tools: PyTorch, Git, Linux, VS Code
 - Generative AI: Diffusion Models, LLMs, RAG Pipelines, AI Agents

Languages: Indonesian (Native), English (IELTS 8.0), Chinese (TOCFL B1)